



SALAS O'BRIEN

3200 WINDY HILL ROAD SE
SUITE 200E
ATLANTA, GEORGIA 30339
PHONE 404-881-5300

Project: Homestead at Hog Mountain; Beltan Properties
Location: Dacula, GA
Description: Six Sisters - CBsmt, E-Facing, ACH50

Date: January 8, 2024
By: JTH / TJH
House Faces: East

These Loads are dependent on the information/assumptions listed below. **Any deviation from these assumptions during construction could have a significant impact on the cooling and heating loads for this house and render the recommended equipment sizing invalid.** If any of these assumptions are incorrect, please let us know immediately so we can re-calculate them. This information will be used in a residential heating and cooling load program based on calculations recommended by ASHRAE, Handbook of Fundamentals. Equipment sizing indicated in this report is valid only upon verification that actual construction is in full compliance with the descriptions below and independent testing of HVAC systems has been performed; such verification and testing shall be performed by Vallus.

BUILDING DATA	
Typical Ceiling Height	10.0 ft
Floor Area	4,187 ft ²
Volume	40,754 ft ³
Percentage Glass	12 %
Number of Bedrooms	6

DESIGN CONDITIONS		
	Summer	Winter
Outdoor	95 °Fdb	18 °Fdb
Indoor	75 °Fdb	72 °Fdb
ΔT	20 °Fdb	54 °Fdb
Range	17.3 °Fdb	

ENERGY CALCULATIONS	
Right Choice HVAC kWh	15024
SEER	14
ACH50	3.0
Mechanical Ventilation	YES
House Outside Air CFM	105

CONSTRUCTION DESCRIPTIONS						
Walls / Partitions	#	U-Value	Description/Notes			
	W1	0.077	Above Grade	2x4	R-13 batt	
	W2	0.048	Basement	Concrete + 2x6	R-19 batt	
	W3	0.053	Attic	2x6	R-19 batt	
Doors	#	U-Value	Description/Notes			
	D1	0.167	Exterior Door		R-6	
	D2	0.167	Garage Door		R-6	
Roofs	#	U-Value	Description/Notes			
	R1	0.026	Flat Ceiling		R-38 blown	
	R2	0.053	Vaulted		R-19 batt	
Floors	#	U-Value	Description/Notes			
	F1	0.077	Framed		R-13 batt	
	F2	0.1	Slab		None	
Glass	#	Glass Type	Windows	U-Value	Internal Shading	Notes
	G1	Double Insulated	All types	0.30	Blinds/Draperies	SHGC 0.18
	G2					
	G3					

LOAD SUMMARY					
		Unit 1 - Whole House			
		Cooling		Heating	
Sensible Load		42690 Btu/hr		50136 Btu/hr	
Minimum Nominal Capacity		4.74 Tons		14.69 kW	
Load Density		883 SF/Ton		3.5 W/SF	
Floor Area Served		4187.0 SF		4187.0 SF	
Mechanical Ventilation		105 CFM		105 CFM	
System Size		5.0 Tons		20.0 kW	
Leaving Air Temperature		57.2 deg F		101.6 deg F	
Installed Capacity		837 SF/Ton		4.8 W/SF	

I understand the information provided herein, and warrant, to the best of my ability and knowledge, and to the full extent of my control as the Contractor, that this house will be built in accordance with the information given above.

Signed _____ Title _____ Date _____

Unit 1				
Terrace Level (Basement) Airflows				
Room	CFM Cool	CFM Heat	CFM Avg	Duct Size
Stairs & Hall	29	55	42	
Bed 5	57	96	76	
Bed 5 Bath	16	34	25	
Bed 5 WIC	3	5	4	
Closet	2	4	3	
Rec Room	89	62	75	
Dining	9	14	11	
Laundry	38	4	21	
Pantry	4	9	7	
Kitchen	135	35	85	
Bed 4 WIC	10	23	16	
Bed 4 Bath	5	8	7	
Bed 4	55	94	74	
Unfinished	54	130	92	

Unit 1				
Size	5.00	Tons		
Airflow	2000	CFM		
Heat kW	20	kW		
First Floor Airflows				
Room	CFM Cool	CFM Heat	CFM Avg	Duct Size
Entry & Living Rm	185	187	186	
Master Bath	49	60	55	
Master WIC (Interior)	8	5	6	
Master Bed	99	89	94	
Master WIC (Rear)	11	21	16	
Breakfast	77	137	107	
Kitchen	179	79	129	
Powder	6	4	5	
Laundry	39	2	21	
Dining	57	103	80	
-	-	-	-	
-	-	-	-	
-	-	-	-	
-	-	-	-	

Unit 1				
Second Floor Airflows				
Room	CFM Cool	CFM Heat	CFM Avg	Duct Size
Stairs (Central)	58	65	62	
Hall	29	42	36	
Bed 2	78	95	87	
Bed 2 Bath	41	43	42	
Bed 2 WIC	12	12	12	
Bed 3 WIC	9	8	8	
Bed 3 Bath	32	45	38	
Bed 3	71	94	83	
-	-	-	-	
Stairs (Garage)	66	92	79	
Kitchen	154	44	99	
Living	165	108	136	
Bath	24	40	32	
Bed	48	50	49	

1438

RC Heating kWh	2679
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Total kWh	4117
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SEER 14

HEATING LOAD CALCULATION																																			
Components				Heating Load		Entire House		Stairs & Hall		Bed 5		Bed 5 Bath		Bed 5 W/C		Closet		Rec Room		Dining		Laundry		Pantry		Kitchen		Bed 4 W/C		Bed 4 Bath		Bed 4		Unfinished	
	Type	Description	$A \times U \times \Delta T$		Factor	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr
Glass	G1		54	0.30	16.2 Btu/hr-sf	74	1195	15	243	29.4	476	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	29.4	476	0	-
	G2		54	0.00	0.0 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	G3		54	0.00	0.0 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
Door	D1		54	0.17	9.0 Btu/hr-sf	5	45	5	45	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	D2		54	0.17	9.0 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	W1	Partition Grade	54	0.08	4.2 Btu/hr-sf	1039	4319	74.2	309	227	945	130	541	0	-	0	-	72.5	301	0	-	0	-	14.2	59	64.2	267	85	353	0	-	223	928	148.3	617
W2	54		0.05	2.6 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	
W3	54		0.05	2.9 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	
W1	27		0.08	2.1 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	
W2	27		0.05	1.3 Btu/hr-sf	609	789	0	-	0	-	0	-	0	-	0	-	72.5	94	0	-	0	-	14.2	18	64.2	83	0	-	0	-	0	-	458.3	594	
R1	54		0.03	1.4 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	
Floor/Roof	R2		54	0.05	2.9 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	F1		54	0.08	4.2 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	F2	Slab	27	0.10	2.7 Btu/hr-sf	1522	4109	148	400	164	443	52	140	29	78	20	54	236	637	76	205	24	65	29	78	104	281	35	95	44	119	160	432	401	1,083
Area				SF	1522		148		164		52		29		20		236		76		24		29		104		35		44		160		401		
Infiltration			35.29 CFM		0.15 Btu/hr	0	2058		200		222		70		39		27		319		103		32		39		141		47		60		216		542
Subtotal (Btu/hr)							12516		1196		2086		751		118		81		1352		308		97		195		771		495		178		2052		2836
Safety Factor				15 %			1877																												
Heat Loss (Btu/hr)							14393																												
Minimum Heater Size (kW)							4.22																												
System Selection			8 kW		Htg. CFM		600		57		100		36		6		4		65		15		5		9		37		24		9		98		136

DX Cooling kWh 2019

By: JTH / TJH

RC Heating kWh	3206
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Level(s): First Floor

Total kWh	5224
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SEER 14

[illegible]

Project:

Location:

Description:

Homestead at Hog Mountain; Beltan Properties

Dacula, GA

Six Sisters - CBsmt, E-Facing, ACH50

Date: January 8, 2024

By: JTH / TJH

Level(s): Second Floor

Square Feet

1233

Volume

12736

SF/Ton

617

SEER

14

Typical Ceiling Height	9	Cooling Size	2	DX Cooling kWh	2235
		Heating Size	2	RC Heating kWh	3447
Kneewall Height	1	Best Case	2	Total kWh	5683

COOLING LOAD CALCULATION						Ceiling Height	20	9	9	9	9	9	9	9	9	9	20	9	9	9	9														
Components		CLTD (°F)	U-Value (Btu/hr-sq-ft-F)	Cooling Load Factor	Entire House		Stairs (Central)		Hall		Bed 2		Bed 2 Bath		Bed 2 WIC		Bed 3 WIC		Bed 3 Bath		Bed 3		Stairs (Garage)		Kitchen		Living		Bath		Bed				
Type	Description				Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr			
Glass	G1 North	Right	(E)	5 Btu/hr-sf	62	327		-		-		-		-		-		-	13.8	72	13.8	72		-	13.4	70		-		-	21.36	112			
	G1 NorthEast		(SE)	7 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-			
	G1 East	Front	(S)	11 Btu/hr-sf	12	135		-		-		-		-		-		-		-		-		-	12.5	135		-		-		-			
	G1 SouthEast		(SW)	12 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-			
	G1 South	Left	(W)	13 Btu/hr-sf	38	501		-		-	13.8	180	13.8	180		-		-		-		-		-		-	10.7	140		-		-			
	G1 SouthWest		(NW)	22 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-			
	G1 West	Back	(N)	24 Btu/hr-sf	62	1507	15	364		-		-	7.5	182		-		-	7.5	182		-		-		-	32	778		-		-			
	G1 NorthWest		(NE)	15 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-			
	G1 Sky Light			38 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-			
D1 Shaded			5 Btu/hr-sf	88	462		-	24	126	32	168		-		-		-		-	32	168		-		-		-		-		-				
Door	D1 Door	21	0.17	3.51 Btu/hr-sf	11	39		-	6.67	23		-		-		-		-		-		-		-	4.45	16		-		-		-			
	D2 Door	55	0.17	9.19 Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-			
Above Grade Walls	W1 North	13	0.08	1.00 Btu/hr-sf	395	396		-	-	-	-	-	-	-	-	-	-	-	46	46	106	106		-	63	63		-	-	87	87	94	94		
	W1 NorthEast	19	0.08	1.46 Btu/hr-sf	0	0		-	-	-	-	-	-	-	-	-	-	-		-		-		-	-	-		-		-		-			
	W1 East	23	0.08	1.77 Btu/hr-sf	369	654		-	59	104	105	185		-		-	-	-	-	-	103	182		-	67	119	36	63		-	-	-			
	W1 SouthEast	21	0.08	1.62 Btu/hr-sf	0	0		-	-	-	-	-		-		-	-	-	-	-		-		-	-	-		-		-		-			
	W1 South	16	0.08	1.23 Btu/hr-sf	382	470		-	-	-	106	131	46	57		-	-	-	-	-	-		-		-	104	128	125	154		-	-	-		
	W1 SouthWest	21	0.08	1.62 Btu/hr-sf	0	0		-	-	-	-	-		-		-	-	-	-	-	-		-		-	-	-		-		-		-		
	W1 West	23	0.08	1.77 Btu/hr-sf	433	767	79	139		-	-	-	35	62	26	46	11	20	74	131		-		-	-	-		-	130	230	78	139		-	
	W1 NorthWest	19	0.08	1.46 Btu/hr-sf	0	0		-	-	-	-	-		-		-	-	-	-	-	-		-		-	-		-		-		-			
	W1 North	13	0.08	1.00 Btu/hr-sf	0	0		-	-	-	-	-		-		-	-	-	-	-	-		-		-	-		-		-		-			
Basement Walls	W1 NorthEast	19	0.08	1.46 Btu/hr-sf	0	0		-	-	-	-	-		-		-	-	-	-	-	-		-		-	-		-		-		-			
	W1 East	23	0.08	1.77 Btu/hr-sf	0	0		-	-	-	-	-		-		-	-	-	-	-	-		-		-	-		-		-		-			
	W1 SouthEast	21	0.08	1.62 Btu/hr-sf	0	0		-	-	-	-	-		-		-	-	-	-	-	-		-		-	-		-		-		-			
	W1 South	16	0.08	1.23 Btu/hr-sf	0	0		-	-	-	-	-		-		-	-	-	-	-	-		-		-	-		-		-		-			
	W1 SouthWest	21	0.08	1.62 Btu/hr-sf	0	0		-	-	-	-	-		-		-	-	-	-	-	-		-		-	-		-		-		-			
	W1 West	23	0.08	1.77 Btu/hr-sf	0	0		-	-	-	-	-		-		-	-	-	-	-	-		-		-	-		-		-		-			
	W1 NorthWest	19	0.08	1.46 Btu/hr-sf	0	0		-	-	-	-	-		-		-	-	-	-	-	-		-		-	-		-		-		-			
	W2 Grade	0	0.05	0.00 Btu/hr-sf	0	0		-	-	-	-	-		-		-	-	-	-	-	-		-		-	-		-		-		-			
	W1 Partition	10	0.08	0.77 Btu/hr-sf	0	0		-	-	-	-	-		-		-	-	-	-	-	-		-		-	-		-		-		-			
Other	W1 Garage	30	0.08	2.31 Btu/hr-sf	126	291		-	-	-	-		-		-	-	-	-	-	-		-		-	-		-		-		-		-		
	W3 Attic	55	0.05	2.92 Btu/hr-sf	289	841	94	273		-	-	-	43	124	26	75	11	33		-	-		-	-	80	232	36	105		-	-	-	-		
	W3 Kneewall	55	0.05	2.92 Btu/hr-sf	0	0		-	-	-	-	-		-		-	-	-	-	-	-		-		-	-		-		-		-			
Roof	R1 Ceiling	55	0.03	1.43 Btu/hr-sf	1226	1753	77	110	84	120	164	235	51	73	31	44	32	46	49	70	162	232		-	65	93	101	144	215	307	68	97	127	182	
	R2 Vaulted	55	0.05	2.92 Btu/hr-sf	0	0		-	-	-	-	-		-		-	-	-	-	-	-		-		-	-		-		-		-			
Floor	F1 Crawl/Uncon.	11	0.08	0.85 Btu/hr-sf	0	0		-	-	-	-	-		-		-	-	-	-	-	-		-		-	-		-		-		-			
	F1 Garage	30	0.08	2.31 Btu/hr-sf	34	79		-	-	-	-	-		-		-	-	-	-	-	-		-		34	79		-		-		-			
Internal	F2 Slab	0	0.10	0.00 Btu/hr-sf	115	0	84		-	-	-	-		-		-	-	-	-	-	-		-		31			-		-		-			
	People			255 Btu/hr	4	1020		-		-	1	255		-		-	-	-	-	-	1	255		-		-		1	255		-	1	255		
Vent	Lights (W)			3.413 Btu/hr-W	620	2116	42	143	42	143	82	280	26	89	16	55	16	55	25	85	81	276		-	33	113	51	174	108	369	34	116	64	218	
	Appliances (W)			3.413 Btu/hr-W	930	3174		-	-	-	-	-		-		-	-	-	-	-	-		-		-	680	2,321	250	853		-	-	-		
Area	Infiltration	32.81	CFM	0.056 Btu/hr-cf		709		93	42		82		26		16		16		25		81			72		51		108		34		64			
	Volume			CF	12736		1680		756	1476	459	279	31	288	441	1458		49	162		0		1300	909		1935	612		1143						
Subtotal (Btu/hr)					15240		1124	558	1516	793	236	169	612	1373		0							1282		2986		3194	473			924				
	Safety Factor			10 %		1524																													
Sensible Cooling Load (Btu/hr)						16764																													
Nominal Tons Required (75% sens. cap.)						1.86																													
System Selection 2 Tons						800		59	29	80	42	12	9	32	72	-		67	157	168	25	49													
HEATING LOAD CALCULATION																																			
Components		CLTD (°F)	U-Value (Btu/hr-sq-ft-F)	Heating Load Factor	Entire House		Stairs		Hall		Bed 2		Bed 2 Bath		Bed 2 WIC		Bed 3 WIC		Bed 3 Bath		Bed 3		0		Stairs		Kitchen		Living		Bath		Bed		
Type	Description				Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	
Glass	G1			54 0.30	16.2 Btu/hr-sf	263	4264	15	243	24	389	45.8	743	21.3	345	0	-	0	-	21.3	345	45.8	743	0	-	25.8	418	0	-	42.7	692	0	-	21.36	346
	G2			54 0.00	0.0 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
	G3			54 0.00	0.0 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
Door	D1			54 0.17	9.0 Btu/hr-sf	11	100	0	-	6.67	60	0	-	0	-	0	-	0	-	0	-	0	-	0	-	4.45	40	0	-	0	-	0	-	0	-
	D2			54 0.17	9.0 Btu/hr-sf	0	0	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-	0	-
Walls	W1			54 0.08	4.																														